

# “I Don’t Know Exactly but I Know a Little”: Exploring Better Responses of Conversational Agents with Insufficient Information

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## ABSTRACT

Despite the increasing presence of conversational agents (CAs) in our daily lives, the lack of information and technology behind them prevents CAs from answering many questions. One of the most typical problems facing conversational user interfaces today is that they often disappoint people by giving the same answer (e.g., “I don’t know”). In this work, we focused on situations in which CAs do not provide a proper answer because of a lack of information. Under these situations, we aimed to find more effective answer strategies for CAs to provide people better user experiences. We tested four different response strategies using different degrees of inferences and information as ground. We found differences in usability and user experience depending on how CAs respond. Our results will help designers understand how people feel about the way CAs respond and create better CA responses in situations where it is difficult to provide accurate answers.

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Natural language interface.

## KEYWORDS

conversational agent(CA), conversational system, voice user interface(VUI), usability, user experience, inference, related information

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## 1 INTRODUCTION

Because voice user interface(VUI) has become popular not only in academia but also among the public, people are more aware of conversational agent (CAs). While an increasing number of people are exposed to CAs, many people still do not use them [5]. One

reason is CAs are not very “conversational.” In fact, they may have only a routine of short answers to simple, specific questions [13]. Moreover, CAs do not fully mimic human conversation, and most of them do not provide satisfactory answers to users. According to a study by [9], user expectations of the CA system are far from the practical realities of use, and users spent a lot of time trying to get the expected task response from CA.

Enough to treat conversation as inferring and making decisions under uncertainty, then imperfect situations always happen during a conversation in the real world [10, 11, 14]. Studies by [3] and [15] showed improved systems for CA by focusing on voice recognition errors. However, even if CA recognizes a voice properly and has a good understanding of what information users need, the lack of sufficient data and query information will bring about the same result. While current CAs try not to undermine the user experience in their own ways, in this case, most of them do not give practical answers that are useful to users.

In human-to-human conversation, people may say “I don’t know,” but they can continue the conversation with some inferences and their grounds. Thus, various kinds of answers arise, unlike when CAs say, “I don’t know,” in a conversation. CAs pursue human-like conversation, unlike other existing interfaces [2], and it is an important design problem that they have no proper answer strategy because of insufficient information, leading to the possibility of building unreliable impressions.

Prior research shows the voice manner (e.g., calm or exuberant) likely has significant effects on psychological and behavior outcomes [12] and how users react according to the quantity of information a CA provides [8], but no attempt has been made to discover response strategies in a circumstance where not enough proper information exists to provide an answer.

In this work, we present a better alternative answer than “I don’t know” when CAs have to give a response but lack information. For that, we attempted to understand users’ perceptions and feelings by creating four CAs that have different response methods when they do not have enough information. The results of a user study of 13 participants showed answers that present relevant reasonable information, even if it does not directly match a user’s question, provides much better usability and a positive image than answers without useful information, such as most current CAs give. Our findings can be used as an effective and useful strategy for designers to decide a CA’s answers in situations where it is difficult to give a user a solid answer.

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**Table 1: Various response ways of common CAs in South Korea (Translation from Korean)**

| CA type                   | An example of response when it fails to find the appropriate answer             | Feature          |
|---------------------------|---------------------------------------------------------------------------------|------------------|
| Clova (Naver)             | “Umm. . . I don’t know.”                                                        | Short and simple |
| Kakao mini (Kakao)        | “I can’t understand what you’re saying.”                                        | Short and simple |
| Siri (Apple)              | “Hmm. . . I don’t know what it is. Is there anything else I can help you with?” | Cooperative      |
| Bixby (Samsung)           | “I’ll study more. There’s no end in learning.”                                  | Friendly         |
| Google Assistant (Google) | Showing the Google search result on the screen                                  | Useful           |

**Table 2: Response setting for Wizard of Oz experiment**

| Answer type | Question type | Response                                                                                                | Inference | Ground of the inference        |
|-------------|---------------|---------------------------------------------------------------------------------------------------------|-----------|--------------------------------|
| A1          | Q1            | “I don’t know.”                                                                                         | X         | X                              |
|             | Q2            | “I don’t know.”                                                                                         |           |                                |
| A2          | Q1            | “I don’t know, but I think so.”                                                                         | O         | X                              |
|             | Q2            | “I don’t know, but I think so.”                                                                         |           |                                |
| A3          | Q1            | “I don’t know, but it’s lunchtime. I think it’s open”                                                   | O         | General fact                   |
|             | Q2            | “I don’t know, but it’s a Korean restaurant. I think it’s good for you.”                                |           |                                |
| A4          | Q1            | “I don’t know, but I think it’s probably because there’s a crowd in front of the restaurant right now.” | O         | Reasonable related information |
|             | Q2            | “I don’t know, but I think it’s probably because there’s a lot of reviews on the website recently.”     |           |                                |

## 2 METHOD

We set up four agents that give simple answers in different ways. We used Wizard of Oz simulation to assume that the voices came from real CAs [6]. We followed the Wizard of Oz method using text-to-speech. The two questions presented were (a) “Is Hoho Korean Restaurant open now?” (Q1, objective question) and (b) “Do you think the menu in Hoho Korean Restaurants will be good?” (Q2, subjective question). They are examples of questions that include objective and subjective facts, respectively, and are simple questions that people often ask in everyday life. We used the two question types to reduce the effect of the result on the characteristics of a particular situation. We set five variations of CA responses: Answer 1 (A1) was “I don’t know.” Answer 2 (A2) was unfounded inference (e.g., “I don’t know, but I think so.”). Answer 3 (A3) was inference based on a general fact (e.g., “I don’t know, but it’s lunchtime. I think it’s open.”). Answer 4 (A4) was inference based on some reasonable related information (e.g., “I don’t know, but I think it’s probably because there’s a crowd in front of the restaurant right now.”). In each type of question, we instructed the participants to ask for information from different restaurants to make the experiment more natural when experiencing four different agents. As such, we conducted the experiment in the form of each participant asking the agent a total eight questions and receiving answers from the agent.

We planned to observe the usability perspective and user experience perspective through the user study. We conducted the survey at the end of each question–answer set using System Usability Scale

(SUS) to examine the usability of each VUI according to different response methods. Developed by Brooke, SUS allows the usability practitioner to assess the usability of a given product or service quickly and easily [1, 4, 16] and is a valid evaluation tool for VUIs [7]. SUS evaluates how a CA’s response is affected by the increased complexity of the VUI system due to its longer response than before. After the survey, we conducted a qualitative assessment by comprehensively asking how the participants felt about each CA through a debriefing interview. We also compared the quantitative and qualitative results.

We invited 13 participants (7 Male, 6 Female; average age 22) who live in South Korea and are native-speaking Koreans. We used Korean for the entire experimental process in this study. For the voice of the CA in this study, we chose a voice available on Google Cloud Text-to-Speech to speak the sentences in the most generic tone, which is ko-KR-Wavenet-C (Language: South Korea; Speed: 1.00, Pitch: 0.00). We recruited a similar number of female and male participants to prevent different effects by gender.

## 3 RESULTS AND FINDINGS

We collected SUS survey results and interview results for 13 participants and interpreted the meaning of the results. Through this process, we found the tendency of CA’s usability. In addition, we compared the usability results and the debriefing interviews according to the CA’s response method (A1–4) and found that the feelings or thoughts that participants had through the CA’s answers even affected the usability of the VUI. In this section, we present an

**Table 3: Descriptive statistic of SUS result data in objective question(Q1) condition**

| Answer type | Average | Standard Deviation | N  |
|-------------|---------|--------------------|----|
| A1          | 52.31   | 18.21              | 13 |
| A2          | 48.85   | 20.50              | 13 |
| A3          | 62.69   | 18.16              | 13 |
| A4          | 76.35   | 9.66               | 13 |

**Table 4: Descriptive statistic of SUS result data in subjective question(Q2) condition**

| Answer type | Average | Standard Deviation | N  |
|-------------|---------|--------------------|----|
| A1          | 60.77   | 12.97              | 13 |
| A2          | 57.50   | 13.62              | 13 |
| A3          | 56.73   | 19.13              | 13 |
| A4          | 69.23   | 24.55              | 13 |

analysis of the usability assessment results for each type of answer and the reasons for their rating.

### 3.1 Trends in System Usability Scale Results

Tables 3 and 4 show participants’ tendency to evaluate the usability of each agent with different response methods. According to [1], with SUS scores less than 50, one can be almost certain that the product will have usability difficulties in the field. In addition, products that are at least passable have SUS scores greater than 70. First, in the Q1 case, A2 (i.e., an answer with unfounded inference) scored less than 50. On the other hand, A4 (i.e., an answer with an inference based on some reasonable related information) had the most distinct usability scores in both Q1 and Q2. In particular, only the average score of A4 exceeded 70 in the Q1 case, and in the Q2 case, it was close to 70. This means only an A4-type CA is passable in the field. The rest of the answers represented values between 50 and 70. Of course, it should be considered that this experiment used one simple question–answer set, not the entire VUI system. Nevertheless, the results show that A4, the answer with an inference with reasonable related information, is significantly more usable than other answers.

### 3.2 CAs that Only Say “I don’t know” Create a Negative Impression

When they heard the answer, “I don’t know,” four participants (P6, P8, P10, and P12) were embarrassed that it was all the CA could provide. Two participants (P10 and P12) were confused about whether the answer was over by saying “I don’t know.”

*“I didn’t know this was the end of the answer. I was expecting the agent to ask me again.” (P8)*

In addition, five participants (P2, P6, P7, P12, and P13) said they would not want to use a CA again if it answered only “I don’t know.” In addition, six participants (P3, P5, P7, P8, P11, and P13) said they felt the CAs were irresponsible and insincere.

*“Why doesn’t the agent bring me a review or something? Why are the agents telling me they don’t know? I don’t think I’m going to use it.” (P2)*

*“Anyway, if my agent keeps answering like this, I don’t want to use this function.” (P6)*

*“It feels like it refused to talk to people because it said too short.” (P3)*

Some said they were annoyed (P5, P6, and P9).

*“Oh, this is annoying, too. I immediately thought it is stupid” (P6)*

*“I was a little upset when it said it didn’t know without giving any reasons.” (P9)*

Overall, participants thought they did not get what they expected from the CAs because they did not look for other information and only answered, “I don’t know.”

### 3.3 An Arbitrary Inference without Reasonable Information Makes a User Feel CAs to be Irresponsible and Inefficient

In the A2 case, which provided unfounded inference, the overwhelming majority of participants said CAs have no responsibility (all participants except P4, P10, and P13). Thus, a CA’s inference that does not provide helpful information gives users a negative image.

*“It was like a nuance that it’s your job, you’re on your own, because it speaks so irresponsibly and unkind.” (P1)*

In addition, some participants (P6, P8, P10, and P12) expressed the view that a clear answer such as A1 (i.e., “I don’t know”) was preferable to a long sentence with meaningless guesses, such as A2 and A3, which also gave no meaningful information.

*“I think so? What are they saying? Are they just tossing it out? This guess feels worse than no answer.” (P6)*

*“I think it is annoying just to say that ‘I don’t know’ so complicatedly.” (P8)*

*“It (A2) was much worse than before (A1). I don’t want to talk long without any information. I think it’s better to just think of it as stupid and end it quickly.” (P12)*

Because a CA usually answers in the same way in many situations where it is difficult to answer, this negative image is likely to continue to undermine usability of CAs.

### 3.4 Providing Specialized Information Increases Sense of Closeness with CAs

In contrast to its usability, when asked about the aspect of closeness, six participants (P4, P6, P7, P10, P11, and P13) said they felt the A3 answer was the best. They felt that this answer was generally like a friend's answer, and that it was not only about the general facts, but also about their personal preference. They felt that the CA was kinder or more self-conscious, even though it once again gave the information they already knew.

*"I think it's the answer from my real friend. This was good." (P6)*

*"It feels like it already knows my taste and told me what I wanted. I think it thought of me." (P11)*

Thus, the amount of new information CAs provide is not the only concern, but also their provision of specialized information is a positive factor that makes people feel closer to the agent. Although SUS survey results did not produce significant differences from this, closeness is another factor CA designers should consider in addition to usability.

### 3.5 Guessing an Answer Using other Related Information had the Highest Usability Score and Positive Evaluation

According to [1], very good products receive SUS scores of 90 or greater. According to the results of each participant's SUS survey, A4 was the only case with four scores (P1, P6 (twice) and P12) of more than 90. Thus, when it is difficult for a CA to give a complete answer, presenting appropriately related information may be a good alternative to enhancing the usability of a CA. All participants except P4, P10, and P13 said they liked A4 because the CA tried to provide a meaningful response to their question.

*"I liked it because I felt like it did its best." (P12)*

They also focused on the fact that the CA provided new information, even if the guess might have been wrong, and they tended not to pay much attention to its accuracy.

*"A well-founded offer is fine. It made me think, 'it tried,' and that it was talking with other objective information." (P9)*

Furthermore, in the subjective question (Q2) case, there was an opinion that it does not matter if the inference is right or wrong.

*"Oh, I like this one. I can decide whether it'll taste good or not because it's just personal taste." (P3)*

Participants suggested that the CA was "uncomfortable" because of the long response without giving a right answer in the A2 and A3 cases, but in the A4 case, most of them did not assess the response in that way and rated the CA most positively, even though it took the longest. Rather, they were satisfied that the CA tried to answer their questions and felt as though it was doing its best to provide useful information. This can be a good opportunity for the users to

have confidence in the agent and to continue using the agent by asking similar questions next time.

*"My CA answers like this, but if other people's CAs are the same as what's on the market right now, I would be very proud of my CA." (P8)*

Some participants focused on the related information that the CA provided and did not realize that the agent did not give them a complete answer, even though it started with "I don't know."

*"What's interesting is I didn't know that it didn't know the answer in this case. Just because it told me something new, and I thought, 'Oh yeah?' ... I just realized it now." (P6)*

Just as human-to-human conversation does not always follow a logical direction, users do not expect from the CA only answers that directly connect to their questions. People consider whether a CA is giving real help or trying to help.

## 4 LIMITATION AND FUTURE WORK

The experiment in this study used a simple question-answer set in each task without real users' needs and situational context. In reality, various situations and complex intents arise when users ask their CAs questions. Future studies can cover situations that are more diverse, classify the related information that CAs can provide, and analyze the features and differences of each category. In addition, we recruited only Korean participants in this study, and Korean cultural characteristics may have influenced the results. Meaningful further research in the age of globalization would examine the distinguishing effects of cultural differences. Also, we set the CA's answer types focusing on the degree of inference and its ground, but response strategies could be extended, such as asking for more clarify on the question. Moreover, Google Assistant currently uses Google's search function to display search results on a smartphone's screen if the CA does not provide an appropriate answer. Thus, further research could be conducted to find the insights and better way to provide relative information in various situations.

## 5 CONCLUSION

In this study, we wanted to find out how a CA's answers can provide a better experience and usability in situations where CAs do not respond properly because of insufficient information. We used the results of quantitative evaluation tool of the usability of CAs with four types of response methods and qualitative analysis of closeness and user experience through debriefing interviews. As a result, when a CA fails to respond fully to a user's question, a user will evaluate the responsibility, sincerity, and level of cooperation the CA tried to provide. We also learned that providing reliable information could significantly improve usability and satisfaction compared to CAs that answer with no side information and only change speech style to signal lack of information, which is how most commercial CAs currently operate. Using our findings to change the way CAs respond will help increase CA usage by building a positive image involving trust, because users will often face uncertain circumstances. CAs not only need to look for information to answer questions directly, but also need to build a system

to rank information that is relevant and discover other information a user who asks a question might require.

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